

Case Study: Artificial Intelligence in Fraud Detection at

Danske Bank

The rise of illegal activities has greatly increased and shifted to digital and mobile platforms. For financial institutions, dealing with these threats manually has become very inefficient. AI has emerged as a solution, making it an essential tool in this area as it helps banks analyze large datasets in real time and recognize complex behaviors. This ultimately assists in identifying complicated behavioral issues and legitimate customer activity.

Danske Bank, a prominent Nordic bank with a 145-year history, successfully implemented AI and deep learning for fraud detection. Before its AI upgrade, Danske Bank depended on an outdated rules-based engine that struggled to keep up with digital banking trends. This old system had only a 40% fraud detection rate and resulted in up to 1,200 false positives each day. Alarming, 99.5% of all flagged transactions were valid. The sheer volume of false alarms required a significant investment of human and financial resources to investigate these

“dead ends.” To modernize its defenses, Danske Bank teamed up with Teradata's Think Big Analytics to launch a cutting-edge AI fraud detection platform.

Technology Overview

To improve its fraud detection abilities, Danske Bank moved from static rules to dynamic Machine Learning (ML) and Deep Learning (DL) methods. The specific technologies and operational frameworks used include: Deep Learning and GPU Integration: The bank incorporated advanced deep learning software optimized with Graphical Processing Unit (GPU) technology. Deep learning models can handle millions of online banking transactions by assessing tens of thousands of latent features at once.

"Champion" Methodology: The fraud detection system uses a continuous competition model. Various deep learning models (the "champion") independently process real-time data streams, learning unique transaction characteristics, like changes in customer location or unusual ATM withdrawal frequencies. When a challenger model outperforms the current champion

mathematically, it replaces it as the new primary algorithm. This ensures the system continuously adapts to changing fraud tactics.

Real-Time Processing Engine: The AI-based platform was designed for speed. It evaluates incoming online, credit card, and mobile transactions in less than 300 milliseconds. This allows the bank to assess a transaction while the customer is still at the checkout counter, offering immediate insights without causing issues for legitimate users.

Interpretation Layer: To address the "black box" issue of deep learning, an interpretation layer was created on top of the machine learning models. This technology gives human analysts clear, understandable reasons for why the AI flagged or blocked particular transactions, which is crucial for human review and regulatory compliance.

Benefits

Implementing an AI analytics solution brought immediate and significant benefits for Danske Bank. By shifting operational decisions from fixed rules to a flexible AI system, the bank greatly enhanced its security and operational efficiency.

Operational and Financial Benefits

By reducing false positives, Danske Bank effectively eliminated the huge administrative backlog that had burdened its compliance officers. The bank redirected 50% of its fraud detection staff from manual data checks to more valuable tasks, like investigating complex, organized financial crime networks. The significant reduction in investigation times saved millions of dollars within the first year, achieving a 100% Return on Investment (ROI) almost immediately.

Challenges

Integrating advanced AI into a highly regulated financial environment is not without difficulties. Danske Bank faced several notable challenges during implementation:

Technical Challenges: The main technical obstacle was merging modern, open-source AI solutions and specialized deep-learning hardware with the bank's older infrastructure. Additionally, detecting fraud is technically challenging due to extreme data imbalance; actual fraud occurs in roughly 1 out of every 100,000 transactions. This meant that the algorithms had

to be carefully trained to find a “needle in a haystack” without overfitting.

Ethical and Regulatory Challenges: In banking, regulations require institutions to explain why a transaction was declined or an account was frozen. The lack of clarity in deep learning algorithms posed serious ethical and regulatory risks. Danske Bank addressed this by creating the “interpretation layer” to ensure analysts could justify the AI's independent decisions.

Operational Challenges: Transitioning trust from human-written rules to machine-driven decisions required a significant cultural change within the compliance team. The bank had to develop a robust tracking system to monitor the models in production constantly, ensuring that the AI could be trusted to operate accurately without ongoing manual checks.

Conclusion

The partnership between Danske Bank and Teradata serves as a key example of AI application in finance. By moving away from outdated rules-based systems in favor of flexible deep-

learning algorithms, Danske Bank tackled the issues of low detection rates and excessive false positives.

Recommendations for other institutions: - Prioritize

Explainability:

Institutions should avoid using “black box” AI. Investing in an interpretation layer is crucial for maintaining regulatory compliance and gaining the trust of human analysts. - Embrace Continuous Learning: Fraudsters constantly change their tactics. Utilizing frameworks like the "Champion" model ensures that fraud defenses are always adjusting to the latest threats. -

Maintain Human Oversight: AI works best as a support tool rather than a complete replacement. While AI can process transactions quickly, complex anomalies still need the judgment of experienced investigators.

Quantitative Improvements

Performance Metric	Legacy Rules-Based System	New AI/Deep Learning System	Net Improvement
Overall Detection Rate	~40% of actual fraud caught	~60% of actual fraud caught	+50% Increase
False Positives (Daily)	Up to 1,200 (99.5% false rate)	Reduced significantly	-60% Decrease
Transaction Scoring Time	High latency / batch processing	< 300 milliseconds	Real-time action

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